SANA ULLAH

M.Phil. Biotechnology • AI-Assisted Molecular Diagnostics • Precision Agriculture

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RESEARCH PROFILE

Biotechnology researcher (M.Phil., CGPA 3.77/4.00) working at the intersection of artificial intelligence and molecular diagnostics in agriculture. Thesis research combined deep learning-based object detection with species-specific PCR and multiplex assay development for real-time monitoring of economically damaging Bactrocera fruit fly species. Seeking a fully funded PhD position that extends this interdisciplinary line of work — applying computational methods to molecular genetics, genomics, and precision agriculture.

EDUCATION

Institution	Degree	Session	CGPA / Marks
University of Agriculture, Peshawar, Pakistan	M.Phil. Biotechnology	2022–2026	3.77 / 4.00
University of Agriculture, Peshawar, Pakistan	B.Sc. (Hons) Biotechnology	2018–2022	3.60 / 4.00
Government College, Peshawar, Pakistan	F.Sc. Pre-Medical	2014–2016	820 / 1100
Frontier Grammar School, Ghari Sherdad, Pakistan	Matriculation (Science)	2012–2014	966 / 1100

RESEARCH EXPERIENCE

M.Phil. Thesis Research — *University of Agriculture, Peshawar* **2022 – 2026**

Artificial Intelligence-Assisted Real-Time Detection of Devastating Fruit Fly Species for Population Monitoring and Effective Control

Supervisor: Dr. Muhammad Sayyar Khan (Ph.D., Germany) • Co-Supervisor: Engr. Dr. Muhammad Hanif (GIKI)

- Collaborated with the Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) to deploy wireless insect traps with high-resolution cameras, capturing real-time fruit fly imagery across a cross-institutional, engineering–biology partnership.
- Designed and validated a YOLO-based deep learning system for real-time detection of four economically damaging *Bactrocera* fruit fly species from trap-camera imagery.
- Developed and validated four species-specific PCR primer pairs and a multiplex PCR assay, confirming molecular identity against a field-collected specimen panel.
- Combined AI-based detection with molecular confirmation into a single population-monitoring workflow, informing evidence-based pest control decisions.

Research Internship — *Plant Tissue Culture & Germplasm Conservation Lab, IBGE, University of Agriculture Peshawar* **2021 – 2022**

Agromorphological and Biochemical Characterization of Advanced Wheat Genotypes under Field Conditions

- Conducted agromorphological assessment of elite wheat genotypes across varying field environments.
- Performed biochemical assays to evaluate genotype performance and stress-tolerance indicators.
- Contributed to germplasm conservation protocols and structured data documentation.

PUBLICATIONS

Ullah, S., Khan, M. S. *Integration of Deep Learning Detection and Species-Specific DNA Markers for Identification of Bactrocera Fruit Flies.* Manuscript in preparation.

Ullah, S., Khan, M. S. *Species-Specific DNA Markers and a Multiplex PCR Assay for Four Bactrocera Fruit Fly Pests in Pakistan.* Manuscript in preparation.

WORK EXPERIENCE

Field Officer — *Dawn Agro Services, Peshawar, Pakistan* **Feb 2021 – Feb 2024**

- Conducted regular field visits and monitored agricultural practices across 20+ farm sites.
- Advised farmers on evidence-based crop management, integrated pest control, and soil health strategies.
- Collected, recorded, and analyzed field performance data to track crop health and yield outcomes.

TECHNICAL & LABORATORY SKILLS

Molecular Biology: DNA extraction (plant/CTAB), PCR, multiplex PCR, gel electrophoresis, Nanodrop spectrophotometry, centrifugation

AI & Computing: Computer vision, applied deep learning (YOLO), Python, MS Office Suite

Bioinformatics: Sequence analysis, primer design principles, NCBI/PubMed literature navigation

Field Research: Agromorphological assessment, biochemical characterization, crop monitoring

General: Scientific writing, data interpretation, research communication

CERTIFICATIONS & TRAINING

Certificate Course in Applied Biomedical AI — OIC-COMSTECH in collaboration with Ningbo University, China (Jul–Aug 2025)

Python Certification — freeCodeCamp.org (2026)

Data Analysis with Python Certification — freeCodeCamp.org (2026)

LANGUAGES

Urdu — Native **Pashto** — Native **English** — Professional Working Proficiency

Medium of instruction (oral and written) for both the B.Sc. (Hons) and M.Phil. degrees was English, certified by the University of Agriculture, Peshawar:

- English Proficiency Certificate — B.Sc. (Hons) Biotechnology, Ref. No. 998/R/UAP (19 Oct 2022).
- English Proficiency Certificate — M.Phil. Biotechnology, Ref. No. 246/R/UAP (31 Mar 2026).

REFERENCES

Dr. Muhammad Sayyar Khan (Ph.D., Germany) — M.Phil. Thesis Supervisor
Associate Professor, Institute of Biotechnology and Genetic Engineering, University of Agriculture Peshawar
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Dr. Quahir Sohail — Academic Referee
Associate Professor, Biochemistry and Plant Biotechnology Division, IBGE, University of Agriculture Peshawar
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